The Cardiovascular System

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The Blood Vessels and Circulation of Blood

- ❑ Structure and Function of Heart
- ❑ Blood Vessels of the Pulmonary Circulation
- ❑ Blood Vessels of the Systemic Circulation
- ❑ The Physiology of Circulation
- ❑ Control of Blood Vessels
- ❑ Regulation of Arterial Pressure

❑ STRUCTURE AND FUNCTION OF HEART

✓ The heart pumps blood

✓ Blood vessels allow blood to circulate to all parts of the body

✓ Functions of the cardiovascular system

- Transport oxygen, nutrients, cell wastes, hormones to and from cells

❖ Anatomy of the Heart

✓ Size of a human fist, weighing less than a pound

✓ Located in the thoracic cavity, between the lungs in the inferior mediastinum

✓ ORIENTATION

- Apex is directed toward left hip and rests on the diaphragm
- Base points toward right shoulder

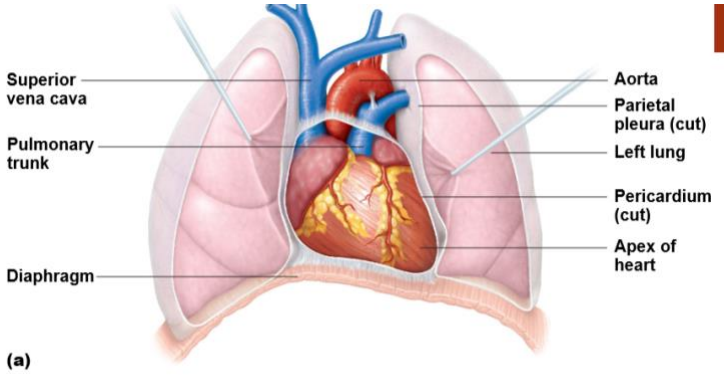
✓ COVERINGS OF THE HEART

Pericardium—a double-walled sac

- Fibrous pericardium is loose and superficial
- Serous membrane is deep to the fibrous pericardium and composed of two layers:
  - Parietal pericardium: outside layer that lines the inner surface of the fibrous pericardium
  - Visceral pericardium: next to heart; also known as the *epicardium*
  - Serous fluid fills the space between the layers of pericardium, called the *pericardial cavity*

✓ WALLS OF THE HEART

1. Epicardium
   - Outside layer; the visceral pericardium
2. Myocardium
   - Middle layer
   - Mostly cardiac muscle
3. Endocardium
   - Inner layer known as *endothelium*



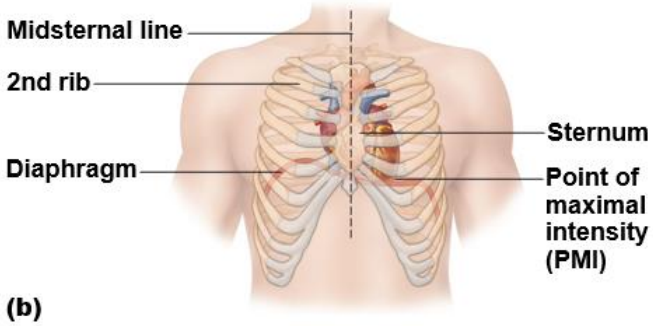
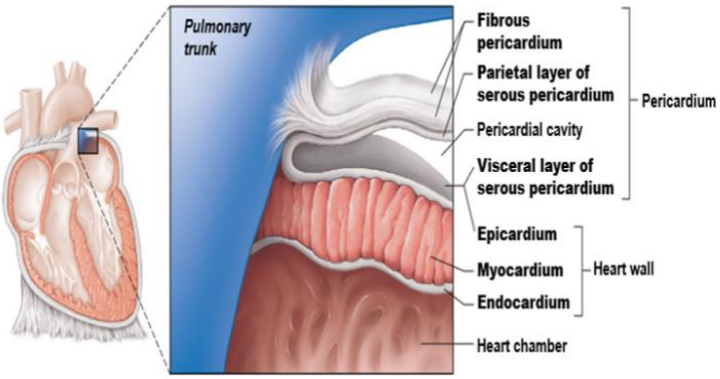


FIGURE 11.1B LOCATION OF THE HEART WITHIN THE THORAX.



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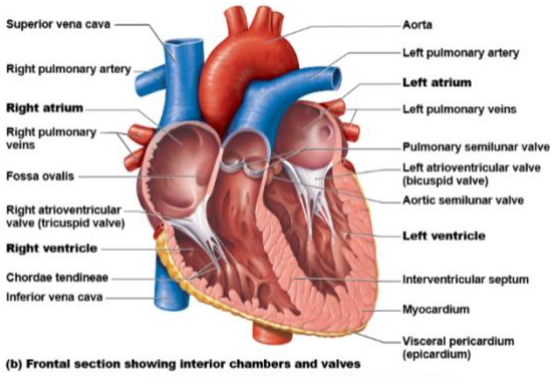
✓ FOUR CHAMBERS OF THE HEART

❖ Atria (right and left)

- Receiving chambers
- Assist with filling the ventricles
- Blood enters under low pressure

❖ Ventricles (right and left)

- Discharging chambers
- Thick-walled pumps of the heart
- During contraction, blood is propelled into circulation



(b) Frontal section showing interior chambers and valves

FIGURE 11.3B GROSS ANATOMY OF THE HEART.

✓ SEPTUM

❖ Interatrial septum

- Separates the two atria longitudinally

❖ Interventricular septum

- Separates the two ventricles longitudinally

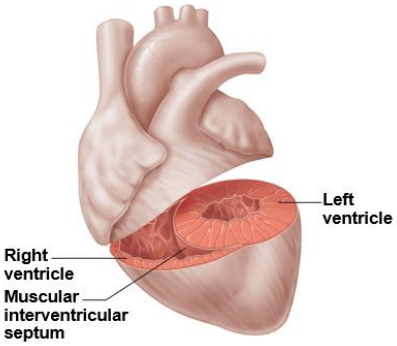


FIGURE 11.5 ANATOMICAL DIFFERENCES IN RIGHT AND LEFT VENTRICLES.

✓ CHAMBERS AND ASSOCIATED GREAT VESSELS

- Heart functions as a double pump
- Arteries carry blood away from the heart
- Veins carry blood toward the heart
- Double pump
  - Right side works as the pulmonary circuit pump
  - Left side works as the systemic circuit pump

❑ BLOOD VESSELS OF THE PULMONARY CIRCULATION

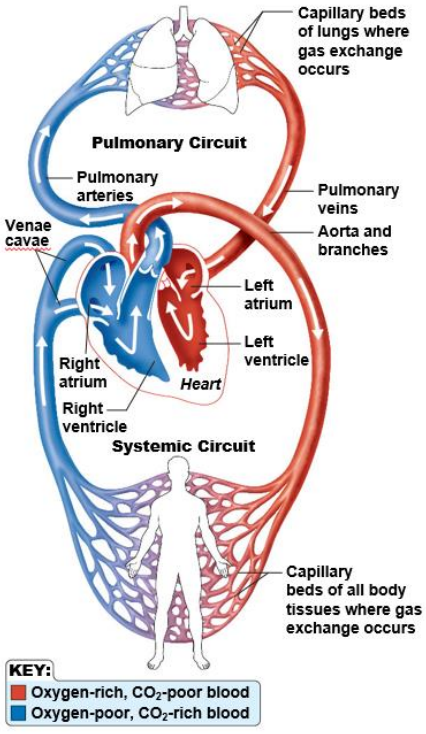
❑ BLOOD VESSELS OF THE SYSTEMIC CIRCULATION

✓ PULMONARY CIRCULATION

- Blood flows from the right side of the heart to the lungs and back to the left side of the heart
- Blood is pumped out of right side through the pulmonary trunk, which splits into pulmonary arteries and takes oxygen-poor blood to lungs
- Oxygen-rich blood returns to the heart from the lungs via pulmonary veins

✓ SYSTEMIC CIRCULATION

- Oxygen-rich blood returned to the left side of the heart is pumped out into the aorta
- Blood circulates to systemic arteries and to all body tissues
- Left ventricle has thicker walls because it pumps blood to the body through the systemic circuit
- Oxygen-poor blood returns to the right atrium via systemic veins, which empty blood into the superior or inferior vena cava

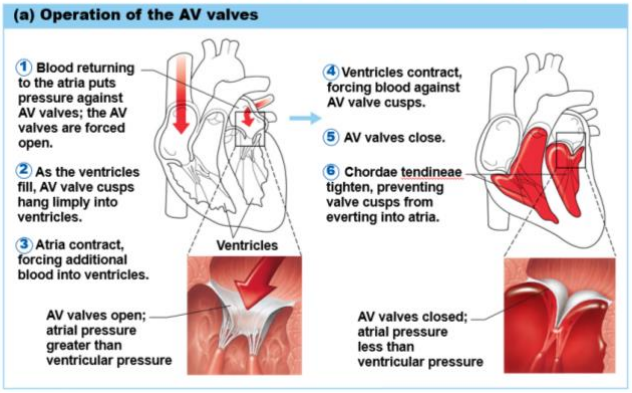


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✓ HEART VALVES

- Allow blood to flow in only one direction, to prevent backflow
  - Atrioventricular (AV) valves—between atria and ventricles
    - Anchored the cusps in place by chordae tendineae to the walls of the ventricles
    - Open during heart relaxation, when blood passively fills the chambers
    - Closed during ventricular contraction
      - Left AV valve: bicuspid (mitral) valve
      - Right AV valve: tricuspid valve
  - Semilunar valves—between ventricle and artery
    - Closed during heart relaxation
    - Open during ventricular contraction
      - Pulmonary semilunar valve
      - Aortic semilunar valve

FIGURE 11.6A OPERATION OF THE HEART VALVES.



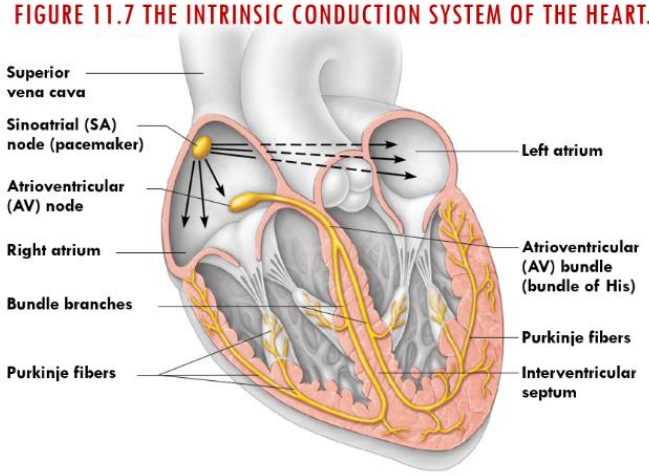
✓ CARDIAC CIRCULATION

- Blood in the heart chambers does not nourish the myocardium
- The heart has its own nourishing circulatory system consisting of:
  - Coronary arteries—branch from the aorta to supply the heart muscle with oxygenated blood
  - Cardiac veins—drain the myocardium of blood
  - Coronary sinus—a large vein on the posterior of the heart; receives blood from cardiac veins
- Blood empties into the right atrium via the coronary sinus

✓ PHYSIOLOGY OF THE HEART

- Cardiac muscle contracts spontaneously and independently of nerve impulses
- Spontaneous contractions occur in a regular and continuous way
- Atrial cells beat 60 times per minute
- Ventricular cells beat 20–40 times per minute
- Two systems regulate heart activity
  - Autonomic nervous system
  - Intrinsic conduction system, or the nodal system
    - Sets the heart rhythm
    - Composed of special nervous tissue
    - Ensures heart muscle depolarization in one direction only (atria to ventricles)
    - Enforces a heart rate of 75 beats per minute
    - Components include:
      - Sinoatrial (SA) node
        - Located in the right atrium
        - Serves as the heart's pacemaker
      - Atrioventricular (AV) node
        - Located at the junction of the atria and ventricles
      - Atrioventricular (AV) bundle (bundle of His) and bundle branches
        - Located in the interventricular septum
      - Purkinje fibers spread within the ventricle wall muscles

FIGURE 11.7 THE INTRINSIC CONDUCTION SYSTEM OF THE HEART.



- Intrinsic conduction system of the heart
  - The sinoatrial node (SA node) starts each heartbeat
  - Impulse spreads through the atria to the AV node
  - Atria contract
  - At the AV node, the impulse is delayed briefly

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- Impulse travels through the AV bundle, bundle branches, and Purkinje fibers
- Ventricles contract; blood is ejected from the heart

✓ CARDIAC CYCLE AND HEART SOUNDS

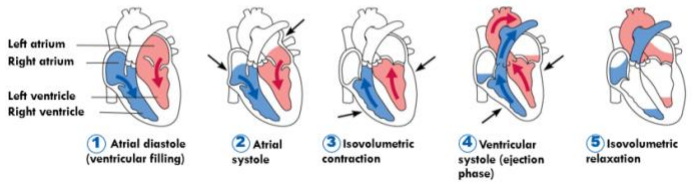
- The *cardiac cycle* refers to one complete heartbeat, in which both atria and ventricles contract and then relax
- Systole = contraction
- Diastole = relaxation
- Average heart rate is approximately 75 beats per minute
- Cardiac cycle length is normally 0.8 second

Atrial Diastole (ventricular filling)	Atrial Systole
▪ Heart is relaxed ▪ Pressure in heart is low ▪ Atrioventricular valves are open ▪ Blood flows passively into the atria and into ventricles ▪ Semilunar valves are closed	▪ Ventricles remain in diastole ▪ Atria contract ▪ Blood is forced into the ventricles to complete ventricular filling

Isovolumetric contraction	Ventricular systole (ejection phase)
▪ Atrial systole ends; ventricular systole begins ▪ Intraventricular pressure rises ▪ AV valves close ▪ For a moment, the ventricles are completely closed chambers	▪ Ventricles continue to contract ▪ Intraventricular pressure now surpasses the pressure in the major arteries leaving the heart ▪ Semilunar valves open ▪ Blood is ejected from the ventricles ▪ Atria are relaxed and filling with blood

Isovolumetric relaxation
▪ Ventricular diastole begins ▪ Pressure falls below that in the major arteries ▪ Semilunar valves close ▪ For another moment, the ventricles are completely closed chambers ▪ When atrial pressure increases above intraventricular pressure, the AV valves open

FIGURE 11.8 SUMMARY OF EVENTS OCCURRING DURING THE CARDIAC CYCLE.



✓ HEART SOUNDS

Lub—longer, louder heart sound caused by the closing of the AV valves
Dup—short, sharp heart sound caused by the closing of the semilunar valves at the end of ventricular systole

✓ PHYSIOLOGY OF THE HEART

❖ Stroke volume (SV)

- Volume of blood pumped by each ventricle in one contraction (each heartbeat)
- Regulation of stroke volume
  - 60 percent of blood in ventricles (about 70 ml) is pumped with each heartbeat
  - Starling's law of the heart
    - The critical factor controlling SV is how much cardiac muscle is stretched
    - The more the cardiac muscle is stretched, the stronger the contraction
- Venous return is the important factor influencing the stretch of heart

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❖ Heart rate (HR)
- Typically 75 beats per minute
- Tachycardia—rapid heart rate, over 100 beats per minute
- Bradycardia—slow heart rate, less than 60 beats per minutes
- Factors Modifying Basic Heart Rate

✓ Neural (ANS) controls
- Sympathetic nervous system speeds heart rate
- Parasympathetic nervous system, primarily vagus nerve fibers, slow and steady the heart rate

✓ Hormones and ions
- Epinephrine and thyroxine speed heart rate
- Excess or lack of calcium, sodium, and potassium ions also modify heart activity

✓ Physical factors
- Age, gender, exercise, body temperature influence heart rate

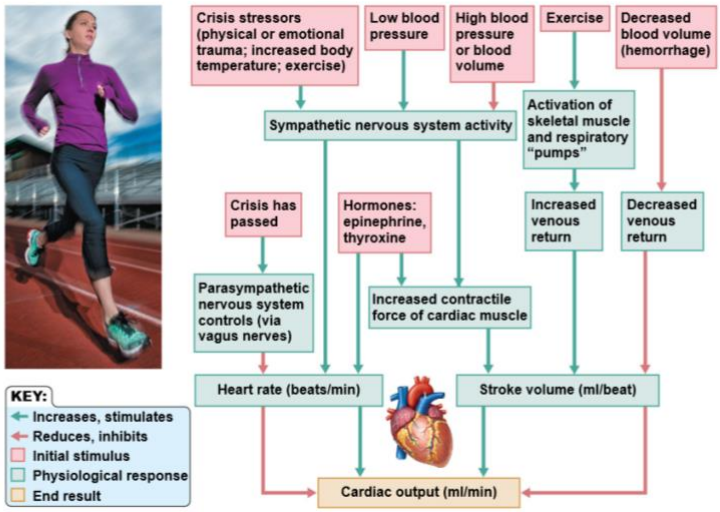


FIGURE 11.9 INFLUENCE OF SELECTED FACTORS ON CARDIAC OUTPUT.

❖ Cardiac output (CO)
- Amount of blood pumped by each side (ventricle) of the heart in 1 minute
- the product of the heart rate (HR) and the stroke volume (SV)
  - CO = HR × SV
  - CO = HR (75 beats/min) × SV (70 ml/beat)
  - CO = 5250 ml/min = 5.25 L/min

❑ CONTROL OF BLOOD VESSELS

✓ Blood vessels form a closed vascular system that transports blood to the tissues and back to the heart

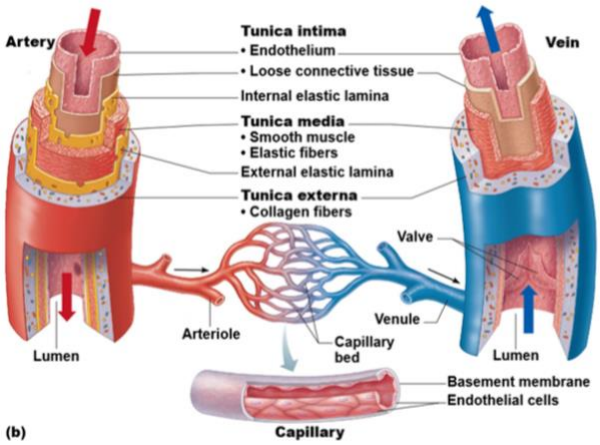
✓ Vessels that carry blood away from the heart
- Arteries and arterioles

✓ Vessels that play a role in exchanges between tissues and blood
- Capillary beds

✓ Vessels that return blood toward the heart
- Venules and veins

✓ Microscopic Anatomy of Blood Vessels

- Three layers (tunics) in blood vessels (except the capillaries)
  - Tunica intima forms a friction-reducing lining
    - Endothelium
  - Tunica media
    - Smooth muscle and elastic tissue
    - Controlled by sympathetic nervous system
  - Tunica externa
    - forms protective outermost covering
    - Mostly fibrous connective tissue
    - Supports and protects the vessel



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True capillaries

- Branch off a terminal arteriole
- Empty directly into a postcapillary venule
- Entrances to capillary beds are guarded by precapillary sphincters

BLOOD VESSELS

MICROSCOPIC ANATOMY: STRUCTURAL DIFFERENCES IN ARTERIES, VEINS, AND CAPILLARIES

Arteries	Veins	Capillaries
▪ have a heavier, stronger, stretchier tunica media than veins to withstand changes in pressure	▪ have a thinner tunica media than arteries and operate under low pressure ▪ have valves to prevent backflow of blood ▪ Lumen of veins is larger than that of arteries	▪ Only one cell layer thick (tunica intima) ▪ Allow for exchanges between blood and tissue ▪ Form networks called *capillary beds* that consist of: ▪ A vascular shunt ▪ True capillaries ▪ Blood flow through a capillary bed is known as *microcirculation*

GROSS ANATOMY OF BLOOD VESSELS

MAJOR ARTERIES OF SYSTEMIC CIRCULATION

- Aorta
  - Largest artery in the body
  - Leaves from the left ventricle of the heart
  - Regions
    - Ascending aorta—leaves the left ventricle
    - Aortic arch—arches to the left
    - Thoracic aorta—travels downward through the thorax
    - Abdominal aorta—passes through the diaphragm into the abdominopelvic cavity

Arterial branches of the ascending aorta

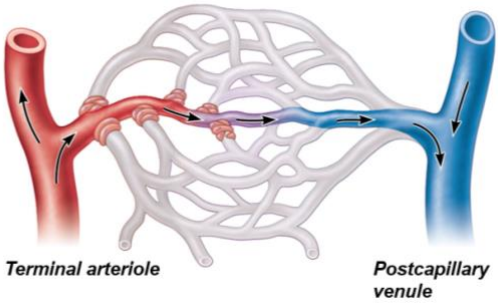
Right and left coronary arteries serve the heart

Arterial branches of the aortic arch

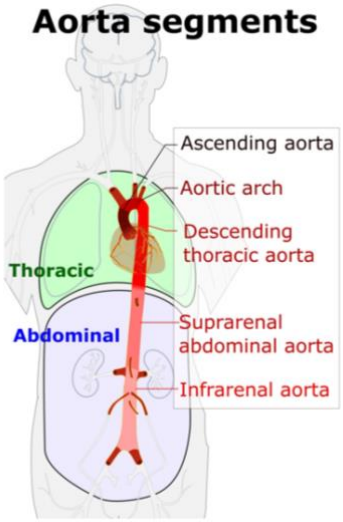
Brachiocephalic trunk splits into the:

- Right common carotid artery
- Right subclavian artery

FIGURE 11.12B ANATOMY OF A CAPILLARY BED.



(b) Sphincters closed; blood flows through vascular shunt.



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Left common carotid artery splits into the:

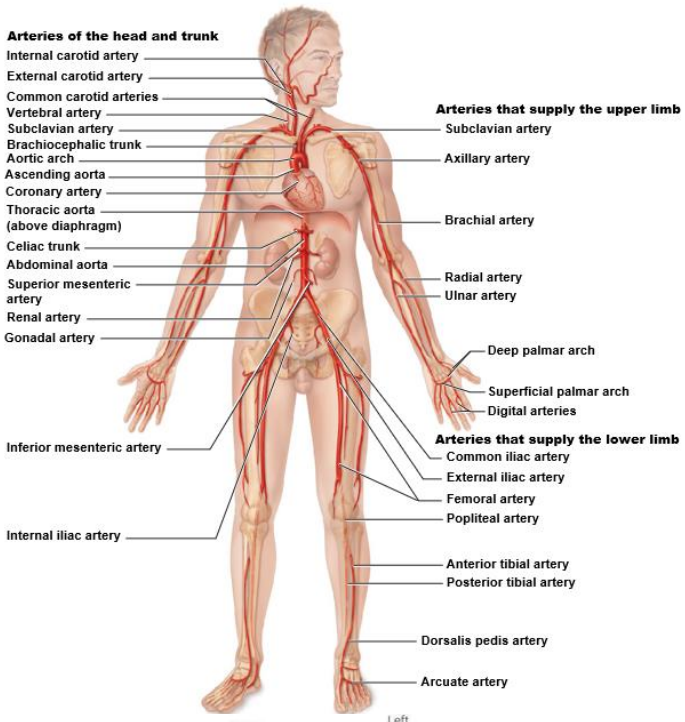
- Left internal and external carotid arteries

Left subclavian artery branches into the:

- Vertebral artery

In the axilla, the subclavian artery becomes the axillary artery → brachial artery → radial and ulnar arteries

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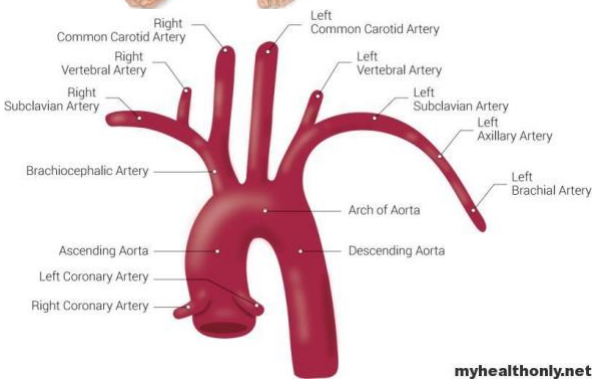


Arterial branches of the thoracic aorta

Intercostal arteries supply the muscles of the thorax wall

Other branches of the thoracic aorta supply the:

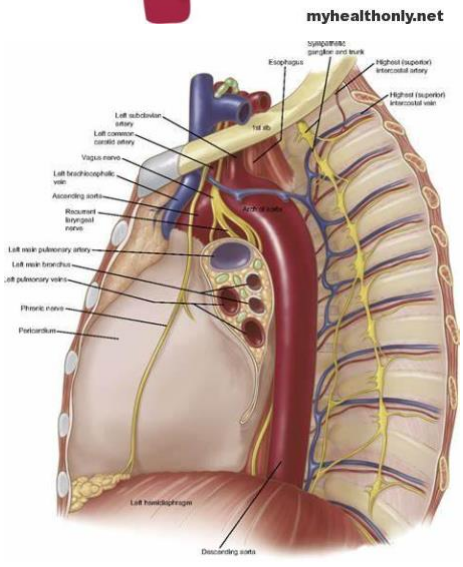
- Lungs (bronchial arteries)
- Esophagus (esophageal arteries)
- Diaphragm (phrenic arteries)



Arterial branches of the abdominal aorta

Celiac trunk is the first branch of the abdominal aorta. Three branches are:

- Left gastric artery (stomach)
- Splenic artery (spleen)
- Common hepatic artery (liver)



Superior mesenteric artery supplies most of the small intestine and first half of the large intestine

Left and right renal arteries (kidney)

Left and right gonadal arteries

- Ovarian arteries in females serve the ovaries
- Testicular arteries in males serve the testes

Lumbar arteries serve muscles of the abdomen and trunk

Inferior mesenteric artery serves the second half of the large intestine

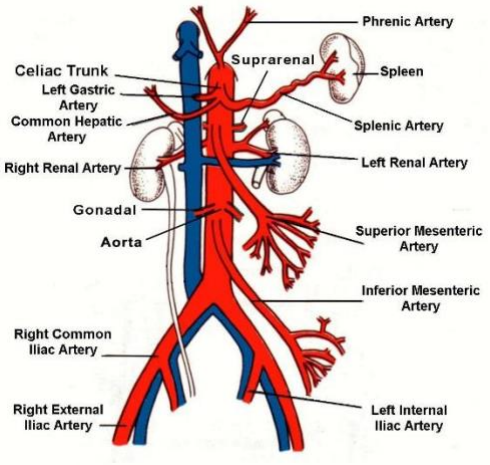
Left and right common iliac arteries are the final branches of the aorta

Internal iliac arteries serve the pelvic organs

External iliac arteries enter the thigh → femoral artery → popliteal artery → anterior and posterior tibial arteries

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MAJOR VEINS OF SYSTEMIC CIRCULATION



Superior vena cava and inferior vena cava
enter the right atrium of the heart
Superior vena cava drains the head and arms
Inferior vena cava drains the lower body

Veins draining into the superior vena cava

Radial and ulnar veins → brachial vein → axillary vein

Cephalic vein drains the lateral aspect of the arm and empties into the axillary vein

Basilic vein drains the medial aspect of the arm and empties into the brachial vein

Basilic and cephalic veins are joined at the median cubital vein (elbow area)

Veins draining into the superior vena cava

Subclavian vein receives:
- Venous blood from the arm via the axillary vein
- Venous blood from skin and muscles via external jugular vein

Vertebral vein drains the posterior part of the head
Internal jugular vein drains the dural sinuses of the brain
Left and right brachiocephalic veins receive venous blood from the:
- Subclavian veins
- Vertebral veins
- Internal jugular veins

Brachiocephalic veins join to form the superior vena cava → right atrium of heart

Azygos vein drains the thorax

Veins draining into the inferior vena cava

Anterior and posterior tibial veins and fibial veins drain the legs

Posterior tibial vein → popliteal vein → femoral vein → external iliac vein

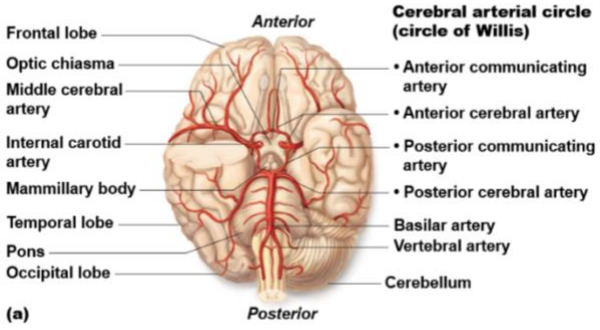
Great saphenous veins (longest veins of the body) receive superficial drainage of the legs
Each common iliac vein (left and right) is formed by the union of the internal and external iliac vein on its own side
Right gonadal vein drains the right ovary in females and right testicle in males
Left gonadal vein empties into the left renal vein
Left and right renal veins drain the kidneys
Hepatic portal vein drains the digestive organs and travels through the liver before it enters systemic circulation
Left and right hepatic veins drain the liver

Arterial supply of the brain and the circle of Willis

Internal carotid arteries divide into:
- Anterior and middle cerebral arteries
  - These arteries supply most of the cerebrum

Vertebral arteries join once within the skull to form the basilar artery
- Basilar artery serves the brain stem and cerebellum

FIGURE 11.15A ARTERIAL SUPPLY OF THE BRAIN.



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Arterial supply of the brain and the circle of Willis

Posterior cerebral arteries form from the division of the basilar artery

These arteries supply the posterior cerebrum

Anterior and posterior blood supplies are united by small communicating arterial branches

Result—complete circle of connecting blood vessels called *cerebral arterial circle*, or *circle of Willis*

Hepatic portal circulation is formed by veins draining the digestive organs, which empty into the hepatic portal vein

Digestive organs

Spleen

Pancreas

Hepatic portal vein carries this blood to the liver, where it is processed before returning to systemic circulation

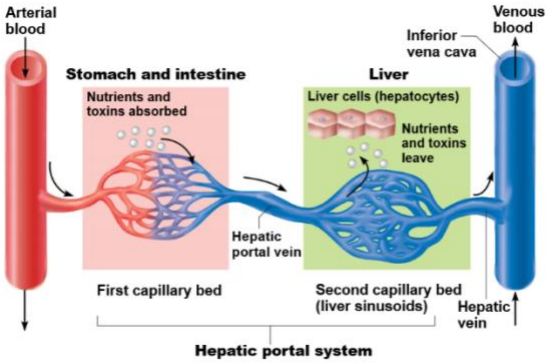
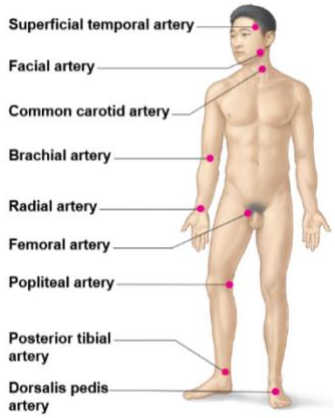


FIGURE 11.16 THE BASIC SCHEME OF THE HEPATIC PORTAL SYSTEM.

❑ REGULATION OF ARTERIAL PRESSURE



✓ Vital signs

Measurements of arterial pulse, blood pressure, respiratory rate, and body temperature

✓ Arterial pulse

- Alternate expansion and recoil of a blood vessel wall (the pressure wave) that occurs as the heart beats
- Monitored at *pressure points* in superficial arteries, where pulse is easily palpated
- Pulse averages 70 to 76 beats per minute at rest, in a healthy person

✓ Blood pressure

- Pressure is high in the arteries, lower in the capillaries, and lowest in the veins
- Measuring blood pressure

Two arterial blood pressures are measured

Systolic—pressure in the arteries at the peak of ventricular contraction

Diastolic—pressure when ventricles relax

Expressed as systolic pressure over diastolic pressure in millimeters of mercury (mm Hg)

For example, 120/80 mm Hg

Auscultatory method is an indirect method of measuring systemic arterial blood pressure, most often in the brachial artery

- Variations in blood pressure

Normal human range is variable

Systolic pressure ranges from 110 to 140 mm Hg

Diastolic pressure ranges from 70 to 80 mm Hg

Hypotension (low blood pressure)

Low systolic (below 100 mm Hg)

Often associated with illness

Acute hypotension is a warning sign for circulatory shock

Hypertension (high blood pressure)

Sustained elevated arterial pressure of 140/90 mm Hg

❖ Developmental Aspects of the Cardiovascular System

In an embryo

✓ The heart develops as a simple tube and pumps blood by week 4 of pregnancy

✓ The heart becomes a four-chambered organ capable of acting as a double pump over the next 3 weeks

✓ Umbilical cord

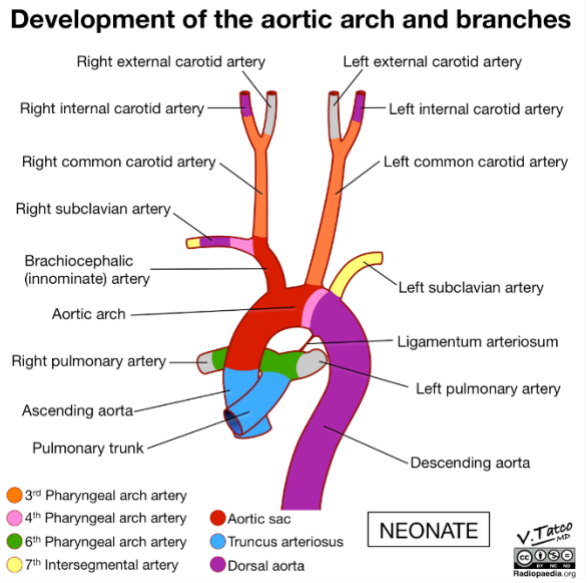
- Carries nutrients and oxygen from maternal blood to fetal blood
- Fetal wastes move from fetal blood to maternal blood
- Houses:

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One umbilical vein, which carries nutrient- and oxygen-rich blood to the fetus
Two umbilical arteries, which carry wastes and carbon dioxide–rich blood from the fetus to placenta

Age-related problems associated with the cardiovascular system include:

- ✓ Weakening of venous valves
- ✓ Varicose veins
- ✓ Progressive arteriosclerosis
- ✓ Hypertension resulting from loss of elasticity of vessels
- ✓ Coronary artery disease resulting from fatty, calcified deposits in the vessels



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